Comprehensive Analysis of the Waveform Dataset: Optimization of kNN and Instance Reduction Techniques

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Abstract

This project investigates supervised machine learning methods on the provided waveform dataset to evaluate classification performance under noisy and overlapping features. The k-Nearest Neighbors (kNN) algorithm is optimized through parameter tuning and bias–variance analysis, followed by instance reduction using Condensed Nearest Neighbor (CNN) and Reduced Nearest Neighbor (RNN). Comparative evaluation with Decision Tree, Random Forest, and Support Vector Machine (SVM) models highlights trade-offs between interpretability, accuracy, and computational cost. The experimental results demonstrate that careful tuning and instance reduction can preserve generalization while improving efficiency. The findings emphasize that properly regularized classical learning methods can achieve near-optimal accuracy even in noisy domains.

1 Introduction

Machine learning algorithms vary in their ability to generalize from noisy, high-dimensional data. The waveform dataset used in this work comprises 5000 samples with 21 features distributed across three classes and includes Gaussian noise to create overlapping boundaries.

The objective of this study is to evaluate the performance of kNN under varying configurations, analyze the effect of instance reduction techniques, and compare its outcomes with Decision Tree, Random Forest, and SVM classifiers. The analysis connects empirical findings with theoretical principles, providing a comprehensive understanding of how data structure, model complexity, and reduction strategies interact to affect predictive accuracy.

2 Experimental Setup

Each instance contains 21 continuous attributes. Preliminary examination confirmed balanced class distribution (33% per class) and moderate feature correlations. All features were standardized using z-score normalization to equalize scale effects under the Euclidean metric.

Before model optimization, data validation and descriptive analysis were performed to verify data integrity, confirm the absence of missing values, and inspect the consistency of feature distributions. These checks ensured that the dataset required no imputation or dimensional adjustments prior to model training.

The data were divided into 4000 training and 1000 testing samples. All experiments were implemented in Python using `scikit-learn` with fixed random seeds to ensure reproducibility.

The experimental procedure included the following stages:

1. Validation and normalization of input data.
2. Determination of the optimal k for kNN using exploratory analysis and cross-validation.
3. Examination of bias–variance dynamics for different training sizes.
4. Application of instance reduction through CNN and RNN.
5. Comparison with Decision Tree, Random Forest, and SVM classifiers.

This sequence establishes a coherent pipeline from preprocessing to comparative evaluation.

2.1 Tuning the Optimal k

Prior to formal optimization, an exploratory sweep of k values was conducted to observe performance trends. This initial evaluation helped narrow the search space to $k \in [50, 100]$ before performing 5-fold cross-validation. Such a two-stage process—empirical exploration followed by systematic validation—ensures both computational efficiency and reliable parameter selection.

The optimization results, illustrated in Fig. 1, show that accuracy increased before stabilizing near $k = 79$, yielding a validation accuracy of approximately 0.843. This behavior reflects the classical bias–variance trade-off: smaller k values exhibit higher variance, whereas larger k values introduce greater bias. The stability at $k = 79$ corresponds with Dudani's distance-weighted variant [3], which improves robustness by assigning greater influence to closer samples. The achieved accuracy closely approaches the theoretical Bayes limit of 0.86, confirming near-optimal tuning.

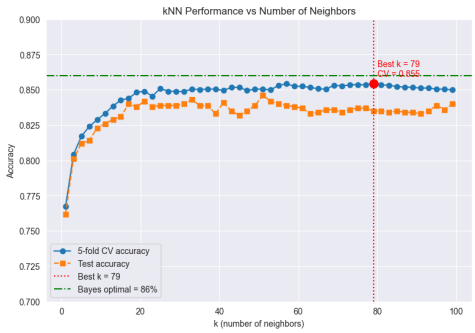


Figure 1: Validation and test accuracy vs. k. The chosen $k = 79$ lies near the elbow point, balancing bias and variance.

2.2 Bias–Variance Behavior

The bias–variance decomposition expresses the expected prediction error as:

$$E\left[(y-\hat{f}(x))^2\right] = \underbrace{(f(x) - E[\hat{f}(x)])^2}_{\text{Bias}^2} + \underbrace{E[(\hat{f}(x) - E[\hat{f}(x)])^2]}_{\text{Variance}} + \sigma^2.$$

In kNN, increasing k reduces variance but increases bias [4, 8, 9]. Smaller k values lead to overfitting by reacting strongly to local noise, while larger k values oversmooth decision boundaries.

The experimental results (Fig. 2) show that models trained on 100 samples display unstable accuracy (high variance), whereas models trained on 3200 samples exhibit smoother, more stable performance. These patterns align with theoretical expectations, as variance decreases roughly proportionally to $1/n$. This confirms that increasing data volume

significantly enhances generalization stability for instance-based models.

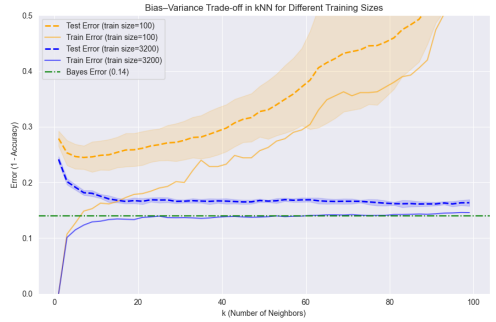


Figure 2: Accuracy for training sizes 100 and 3200. Larger datasets stabilize generalization and reduce variance.

2.3 Instance Reduction Techniques

The computational cost and memory requirements of kNN increase with dataset size. Instance reduction methods, such as CNN and RNN, mitigate this by pruning redundant data while preserving accuracy.

2.3.1 Condensed Nearest Neighbor (CNN)

CNN [1] constructs a minimal subset that correctly classifies remaining instances. It reduced the dataset by 52.75% with a 3.3% accuracy drop (Table 1). Although efficient, the method occasionally discards boundary samples, reducing decision precision [5].

Table 1: CNN reduction performance (time in seconds).

Orig	Reduced	Rate	Time	Acc.B	Acc.A	Δ
4000	2110	52.75%	108.8	0.762	0.729	0.033

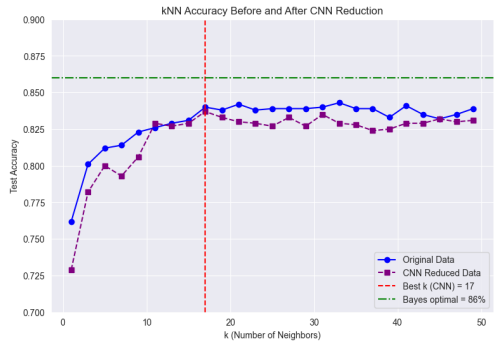


Figure 3: Accuracy before and after reduction using CNN.

As illustrated in Fig. 3, CNN nearly halves the training set while retaining most of the predictive performance. The observed decline in accuracy suggests that boundary points important for class separation were removed, reflecting a trade-off between dataset compactness and decision fidelity.

2.3.2 Reduced Nearest Neighbor (RNN)

RNN [2] iteratively removes non-essential samples while preserving classification performance. It produced a 40% reduction in data with only a 0.5% loss in accuracy (Table 2). This approach maintains boundary structure more effectively and achieves stable results, suitable for real-time systems.

Table 2: RNN reduction performance (time in seconds).

Orig	Reduced	Rate	Time	Acc.B	Acc.A	Δ
4000	2400	40.00%	115.2	0.762	0.757	0.005

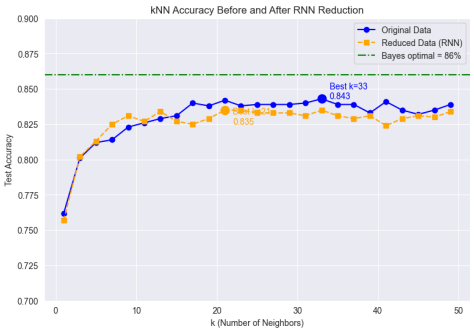


Figure 4: Accuracy before and after reduction using RNN.

Figure 4 shows that RNN maintains nearly identical accuracy despite substantial reduction. The minimal difference between pre- and post-reduction accuracy confirms that RNN effectively eliminates redundant instances while preserving decision boundaries. In comparison, RNN provides a more balanced compromise between efficiency and stability than CNN.

2.4 Comparison with Other Classifiers

The optimized kNN and its reduced variants were evaluated against Decision Tree, Random Forest, and SVM (RBF). SVM achieved the highest accuracy, followed by Random Forest and kNN. The results, shown in Fig. 5, were obtained through direct test set evaluation.

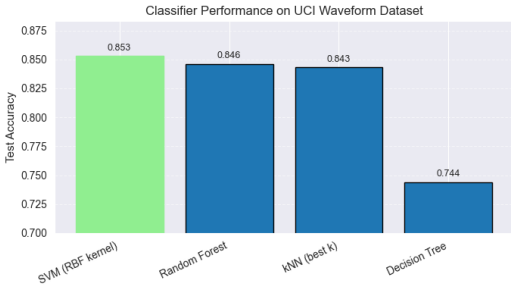


Figure 5: Comparison of test accuracies among classifiers. SVM outperforms, followed by Random Forest and kNN.

These results indicate that ensemble and kernel-based models outperform instance-based methods in handling complex decision boundaries. However, the simplicity and interpretability of kNN make it favorable in scenarios where model transparency or incremental retraining is required.

3 Analysis of Results

The comprehensive experimental evaluation provides several insights into the generalization behavior and computational efficiency of kNN and its reduced variants.

The kNN classifier, when properly tuned, demonstrated remarkable stability and accuracy across multiple configurations. The optimal neighborhood size of $k = 79$ yielded

a validation accuracy of approximately 0.843, closely approaching the theoretical Bayes limit of 0.86. This confirms that even a simple, non-parametric model can achieve near-optimal classification performance when parameters are tuned appropriately. The gradual improvement followed by saturation in accuracy with increasing k reflects the well-established bias–variance trade-off, where small k values amplify variance by overfitting local noise, and large k values increase bias by oversmoothing class boundaries. The bias–variance plots further validated this principle, showing that model variance decreases as the training set size grows—highlighting the stabilizing role of larger data volumes in instance-based learning.

From the reduction perspective, CNN and RNN offered distinct yet complementary advantages. CNN achieved a substantial 52.75% reduction in training data, cutting computational cost almost in half. However, the loss of some boundary-defining samples led to a modest decline in accuracy (3.3%), indicating that aggressive condensation can distort complex decision surfaces. RNN, on the other hand, prioritized stability by iteratively pruning only redundant instances, resulting in a 40% reduction with negligible performance loss (0.5%). The preserved class boundaries and smoother accuracy curves after RNN reduction indicate stronger robustness against noise and a more faithful representation of decision structures. These outcomes empirically reinforce the theoretical premise that controlled instance reduction can improve inference efficiency without compromising predictive reliability.

Comparisons with other classifiers further contextualized the findings. Random Forest and SVM both slightly outperformed kNN, with accuracies around 0.85, consistent with prior literature [6]. The superior performance of SVM arises from its ability to define non-linear decision boundaries in higher-dimensional feature spaces through kernel transformations. Random Forest, by aggregating multiple decorrelated decision trees, effectively reduces overfitting and variance, leading to better generalization on noisy data. Nevertheless, kNN maintained competitive performance despite its simplicity and lack of explicit model training, highlighting its robustness for low-dimensional, moderately noisy tasks.

Overall, the experiments confirm several key insights:

- Bias–variance dynamics in kNN behave consistently with theoretical expectations, with larger datasets stabilizing performance.
- Instance reduction methods like RNN can enhance computational efficiency while maintaining generalization.
- Ensemble and kernel-based classifiers outperform kNN in complex boundary scenarios but at a higher computational cost.

The analysis underscores that, although modern ensemble or kernel methods achieve marginally higher accuracy, the interpretability, incremental learning capability, and parameter simplicity of kNN remain significant advantages. When properly tuned and reduced, kNN provides a strong balance between performance and efficiency, making it a valuable benchmark in both educational and practical machine learning contexts.

4 Conclusion

A detailed evaluation of kNN and its reduction variants was conducted on the waveform dataset. The optimized kNN model reached 0.843 accuracy, approaching the Bayes limit. Bias–variance dynamics were confirmed empirically, and both CNN and RNN improved computational efficiency with minor trade-offs. RNN was found to preserve decision boundaries more effectively. Comparisons with ensemble and kernel methods confirmed their performance advantage but also highlighted the continuing relevance of well-tuned instance-based learning. Future research may include adaptive distance metrics, hybrid ensemble–kNN methods, or dimensionality reduction to enhance scalability for large real-world datasets.

References

[1] P. Hart, "The condensed nearest neighbor rule," IEEE Trans. Information Theory, 1970.

[2] D. Aha, "Incremental learning with reduced instance bases," Machine Learning, 1989.

[3] S. Dudani, "The distance-weighted k-nearest neighbor rule," IEEE Trans. Syst. Man Cybern., 1976.

[4] P. Domingos, "A unified bias–variance decomposition," in Proc. 17th Int. Conf. Machine Learning, 2000.

[5] D. Wilson, T. Martinez, "Reduction techniques for instance-based learning algorithms," Machine Learning, 2000.

[6] L. Breiman, "Random forests," Machine Learning, 2001.

[7] G. Brown et al., *Bias/Variance is not the same as Approximation/Estimation*, Manchester, 2024.

[8] A. Gupta, "Derivation of bias–variance decomposition expression for k-nearest neighbor regression," Stats StackExchange, 2016.

[9] V. M. Nedel'ko, "On the properties of bias-variance decomposition for kNN regression," Bull. Irkutsk State Univ., 2023.

[10] "Bias–variance tradeoff," Wikipedia (accessed 2025).